

EDUCATION

MS in Computer Software Engineering Systems
Northeastern University, Boston, MA

Expected August 2026
GPA: 3.6/4.0

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C, HTML/CSS

Backend & Systems: FastAPI, Flask, Spring Boot, Node.js, REST APIs, GraphQL, microservices, async processing, multithreading, OOP, data structures & algorithms

Databases: PostgreSQL, Redis, MongoDB, MySQL — schema design, indexing, query optimization, caching

Cloud & DevOps: AWS (EC2, Lambda, S3, RDS, ECS, CloudWatch), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, Linux/Unix

AI/ML: PyTorch, TensorFlow, Scikit-learn, LangChain, RAG (Pinecone, pgvector), LLM evaluation

PROFESSIONAL EXPERIENCE

StageZero Health

January 2026 - May 2026

Software Engineering Intern

Boston, MA

React Native, Next.js, TypeScript, Firebase, Cloud Run, AWS, Pinecone, pdfplumber, Tesseract, LangChain

- Reduced onboarding drop-off by 35% by shipping a cancer risk app with **React Native**, **Next.js**, and Google OAuth; guided users through a risk questionnaire to classify them into personalized cancer risk personas.
- Built a **RAG pipeline** with **Tesseract OCR** parsing 100+ scanned medical reports at 95%+ accuracy; extracted insights fed into a risk model to generate personalized breast cancer risk profiles.
- Cut document query time by 40% with a **Pinecone** semantic search microservice over patient records on AWS/GCP.
- Improved deployment reliability by 30% via **Firebase**/Cloud Run with CI/CD; Sentry cut incidents by 25%.

Accenture

July 2021 - June 2024

Custom Software Engineering Analyst

Hyderabad, India

Python, Java, Tableau, ChatGPT API, Pandas, NumPy, Scikit-learn, NLTK, SQL

- Migrated 500TB+ financial data (Bank of America) via **Python/Java ETL pipelines**; flagged 2,000+ issues across 10M+ transactions, boosting reliability by 35%; promoted in 7 months.
- Improved deployment reliability by 20% (Travelers Insurance) with **Tableau dashboards** across 150+ apps.
- Boosted SLA compliance by 30% (DTE Energy) via **Python NLP** ticketing system routing 10K+ weekly tickets at 92% accuracy.
- Improved chatbot accuracy by 25% (Momentum) via **ChatGPT API** and sentiment analysis; deployed to 5 teams.

TECHNICAL PROJECTS

AI World — Generative-Agent Simulation (in progress)

June 2026 - Present

Python, FastAPI, PostgreSQL/pgvector, Ollama

- Built a persistent simulation of ≤ 7 **LLM agents** each running a full cognitive loop (perceive → retrieve → decide → act → reflect) per tick, with population capped via birth/death lifecycle.
- Persisted all world state and agent memories in **PostgreSQL/pgvector**; simulation survives full Docker restarts with complete memory and lineage intact across sessions.
- Implemented emergent behaviors including conversation, conflict/death, teaching, and **reproduction** with LLM-generated child personas and blended inherited traits.

Distributed Super Resolution Framework

January 2026

PyTorch, Multi-GPU

- Built an image super-resolution system combining **ESRGAN** and two **VQGAN-Transformer** architectures, benchmarking perceptual quality across models.
- Optimized distributed training (**DDP/FSDP**) achieving 1.7x speedup and 40% memory reduction via gradient accumulation and mixed-precision.

ACHIEVEMENTS

1st Place — Precision Neuroscience BCI Hackathon (\$1,000 Prize)

2026

Achieved 92% accuracy on 1024-channel ECoG classification with <5ms latency and <1MB model size using **EEGNet** with PCA (1024→32 channels) for real-time edge deployment in 48 hours.